

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1909

Mickens 345 kV
December 01, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1909. The Interconnection Customer proposes a 500.0 MW battery energy storage facility interconnecting at the Mickens 345 kV substation in the Virginia area. The request was received on October 15, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 647 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,356,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: NE-HOBBS-HARBOR BEACH 138 KV L
Contingency: MIRA VISTA-YAREN 345 KV
Loading: 123.3% (Rating: 534 MVA)
DFAX: 11.6%
Pre-project loading: 112.4%
2. Facility: NO MANS LAND-NE-HOBBS 138 KV
Contingency: YARNEN-MIRA VISTA 345 KV
Loading: 120.9% (Rating: 518 MVA)
DFAX: 11.6%
Pre-project loading: 109.7%
3. Facility: KEWAUNEE-TYLER 138 KV
Contingency: N TOWER TAP-MCCARTNEY 345 KV
Loading: 108.1% (Rating: 227 MVA)
DFAX: 7.0%
Pre-project loading: 92.8%
4. Facility: SYL LASKIN-MITCHELL DAM 138 KV
Contingency: HAGGIN-MITCHELL DAM 345 KV Line
Loading: 106.8% (Rating: 350 MVA)
DFAX: 7.0%
Pre-project loading: 96.8%
5. Facility: NEWPORT NEWS-MICKENS 345 KV
Contingency: IGO-MICKENS 345 KV
Loading: 106.4% (Rating: 1,513 MVA)
DFAX: 33.9%
Pre-project loading: 95.2%

6. Facility: WING LANE SUBSTATION-SECO 138 KV
Contingency: MITCELL DAM-SKAGGS ISLAND JCT 2 345 KV
Loading: 105.5% (Rating: 573 MVA)
DFAX: 9.4%
Pre-project loading: 97.3%
7. Facility: IGO-MICKENS 345 KV
Contingency: NEWPORTNEWS SUBSTATION-MICKENS 345 KV
Loading: 105.1% (Rating: 1,399 MVA)
DFAX: 30.1%
Pre-project loading: 94.4%
8. Facility: REDBED PLAINS WIND FARM-FRUITRIDGE & S. LAND PARK 138 KV
Contingency: MITCHEL DAM-SKAGGS ISLAND JUNCTION 2 345 KV
Loading: 104.5% (Rating: 444 MVA)
DFAX: 5.9%
Pre-project loading: 97.9%
9. Facility: NORTH TOWER TAP SUBSTATION-HAGGIN 345 KV
Contingency: MECKLENBURG POWER STA SUBSTATION-BEAR 345 KV L
Loading: 104.5% (Rating: 1,742 MVA)
DFAX: 23.9%
Pre-project loading: 97.6%
10. Facility: MAGNOLIA-COSMAR 138 KV
Contingency: CALIENTE-MAGNOLIA 138 KV line
Loading: 104.0% (Rating: 274 MVA)
DFAX: 8.1%
Pre-project loading: 89.2%
11. Facility: MCCARTNEY-N TOWER TAP 345 KV
Contingency: BEAR-MECKLENBURG POWER STA 345 KV
Loading: 103.5% (Rating: 1,698 MVA)
DFAX: 27.4%
Pre-project loading: 95.5%
12. Facility: DEEN PT-GOLDFIELD 138 KV
Contingency: MCCARTNEY-N TOWER TAP 345 KV
Loading: 103.0% (Rating: 266 MVA)
DFAX: 7.0%
Pre-project loading: 89.9%
13. Facility: CEDAR FALLS WI-OYSTER RIV 138 KV
Contingency: N TOWER TAP-HAGGIN 345 KV
Loading: 103.0% (Rating: 212 MVA)
DFAX: 5.0%
Pre-project loading: 91.2%
14. Facility: MITCHELL DAM 345 kV/138 kV Auto
Contingency: MITCHELL DAM-HAGGIN 345 KV
Loading: 103.0% (Rating: 363 MVA)
DFAX: 7.0%
Pre-project loading: 93.3%
15. Facility: BEAR ISLAND 345 kV/1 kV Transformer Bank
Contingency: CHANCELLOR ROW-BEAR ISLAND 345 KV
Loading: 102.7% (Rating: 588 MVA)
DFAX: 7.2%
Pre-project loading: 96.6%
16. Facility: MCCARTNEY-ROANOKE RAPIDS 345 KV line
Contingency: MCCARTNEY-GASTON 345 KV

Loading: 102.7% (Rating: 1,852 MVA)

DFAX: 14.9%

Pre-project loading: 98.7%

17. Facility: CARLIN-FRUITRIDGE & S. LND PARK 138 KV
Contingency: SKAGGS ISLAND JCT 2-MITCHELL DAM 345 KV
Loading: 102.7% (Rating: 495 MVA)
DFAX: 5.9%
Pre-project loading: 96.7%

18. Facility: BEAR SUBSTATION-CLOVER 345 KV
Contingency: BEAR-MECKLENBURG POWER STA SUBSTATION 345 KV line
Loading: 102.6% (Rating: 4,332 MVA)
DFAX: 7.2%
Pre-project loading: 101.8%

19. Facility: FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV
Contingency: BIRCHWOOD-LEAVITT 345 KV line
Loading: 102.4% (Rating: 3,520 MVA)
DFAX: 12.7%
Pre-project loading: 100.6%

20. Facility: OYSTER RIVER-HARVELL 138 KV line
Contingency: NORTH TOWER TAP-HAGGIN 345 KV
Loading: 102.2% (Rating: 191 MVA)
DFAX: 5.0%
Pre-project loading: 89.1%

21. Facility: HARVAD-SECO 138 KV line
Contingency: MITCHELL DAM SUBSTATION-SKAGGS ISLAND JCT 2 SUBSTATION 345 KV
Loading: 102.1% (Rating: 705 MVA)
DFAX: 9.4%
Pre-project loading: 95.4%

22. Facility: LOUISA GENERATION FACILITY-N ANNA 345 KV
Contingency: CARABELLE-LOUISA GENERATION FACILITY 345 KV line
Loading: 100.7% (Rating: 3,019 MVA)
DFAX: 5.9%
Pre-project loading: 99.7%

23. Facility: WESTCOAST CELLUFIBRE-LEAVITT 138 KV
Contingency: SKAGGS ISLAND JCT 2 SUBSTATION-MITCHELL DAM 345 KV L
Loading: 100.6% (Rating: 561 MVA)
DFAX: 7.7%
Pre-project loading: 93.7%

24. Facility: LEAVITT-WESTCOAST CELLUFIBRE 138 KV Circuit 2
Contingency: MITCHELL DAM-SKAGGS ISLAND JCT 2 345 KV
Loading: 100.6% (Rating: 561 MVA)
DFAX: 7.7%
Pre-project loading: 93.7%

25. Facility: PLAZA-BIG HANAFORD SUBSTATION 138 KV
Contingency: MITCHELL DAM-SKAGGS ISLAND JCT 2 345 KV
Loading: 100.3% (Rating: 818 MVA)
DFAX: 5.1%
Pre-project loading: 97.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NE-HOBBS-HARBOR BEACH 138 KV L	Rebuild line with higher-capacity conductor	\$5,365,000	9% (\$467,000)
NO MANS LAND-NE-HOBBS 138 KV	Rebuild line with higher-capacity conductor	\$2,212,000	9% (\$192,000)
TYLER-KEWAUNEE 138 KV line	Reconductor line	\$3,113,000	5% (\$165,000)
MITCHELL DAM-SYL LASKIN 138 KV line	Reconductor line	\$722,000	5% (\$37,000)
NEWPORT NEWS-MICKENS 345 KV	Reconductor line	\$23,174,000	18% (\$4,046,000)
WING LANE SUBSTATION-SECO 138 KV	Reconductor line	\$564,000	5% (\$29,000)
IGO-MICKENS 345 KV	Reconductor line	\$13,448,000	19% (\$2,514,000)
REDBED PLAINS WIND FARM-FRUITRIDGE & S. LAND PARK 138 KV	Reconductor line	\$2,270,000	4% (\$93,000)
N TOWER TAP-HAGGIN 345 KV L	Reconductor line	\$13,236,000	7% (\$944,000)
COSMAR SUBSTATION-MAGNOLIA SUBSTATION 138 KV	Reconductor line	\$4,889,000	12% (\$592,000)
N TOWER TAP-MCCARTNEY 345 KV	Reconductor line	\$30,352,000	7% (\$2,163,000)
DEEN PT-GOLDFIELD 138 KV	Reconductor line	\$562,000	5% (\$30,000)
OYSTER RIV-CEDAR FALLS WI 138 KV	Reconductor line	\$1,143,000	8% (\$92,000)
MITCHELL DAM 345-138 KV TX	Replace transformer with higher-rated unit	\$20,948,000	5% (\$1,081,000)
BEAR ISLAND 345 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$19,534,000	4% (\$836,000)
MCCARTNEY-ROANOKE RAPIDS 345 KV line	Reconductor line	\$15,420,000	6% (\$917,000)
CARLIN-FRUITRIDGE & S. LND PARK 138 KV	Reconductor line	\$755,000	4% (\$31,000)
BEAR SUBSTATION-CLOVER 345 KV	Reconductor line	\$31,686,000	6% (\$1,817,000)
FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV Line	Reconductor line	\$33,683,000	5% (\$1,591,000)
OYSTER RIV-HARVELL 138 KV	Reconductor line	\$603,000	8% (\$49,000)
HARVAD-SECO 138 KV line	Reconductor line	\$1,847,000	5% (\$96,000)
LOUISA GENERATION FACILITY-N ANNA 345 KV	Reconductor line	\$32,328,000	4% (\$1,388,000)
WESTCOAST CELLUFIBRE-LEAVITT 138 KV	Reconductor line	\$510,000	4% (\$20,000)
LEAVITT-WESTCOAST CELLUFIBRE 138 KV Circuit 2	Reconductor line	\$663,000	4% (\$26,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PLAZA-BIG HANAFORD SUBSTATION 138 KV	Reconductor line	\$4,827,000	3% (\$140,000)
	Total allocated to project		\$19,356,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.